Maximum Mark: 10 **Quiz-1** **Time: 55 minutes**

1. (points) [1.5+1.5+1]
   (a) Estimate the location at which the radial probability density is a maximum for the ground state of the Hydrogen atom.
   (b) Estimate the expectation value for the radial coordinate in the ground state of the Hydrogen atom.
   (c) Are the above two results same? Interpret these in terms of the measurement of location of the electron in the atom.

   $\psi_{100} = \frac{1}{\sqrt{\pi}}(\frac{1}{a_0})^{3/2}e^{-r/a_0}$

2. (2 points) Using the Heisenberg's uncertainty principle, obtain an estimate of the size and energy of an H atom (use Logical assumptions if necessary)?

3. (2 points) Estimate the 2nd order correction to E i.e. $E_k^{(2)}$ using time independent perturbation approach.

4. (2 points) [0.5*4]
   (a) Plot $V_{eff} = \frac{-Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)\hbar^2}{2\mu r^2}$ for at least 3 different values of l.
   (b) Plot the radial distribution function (probability per unit length) for
   $R_{10} = 2\left(\frac{1}{a_0}\right)^{3/2} e^{(-r/a_0)}$,
   $R_{20} = 2\left(\frac{1}{2a_0}\right)^{3/2}\left(1 - \frac{r}{2a_0}\right) e^{(-r/2a_0)}$,
   $R_{31} = \frac{4\sqrt{2}}{9}\left(\frac{1}{3a_0}\right)^{3/2}\left(1 - \frac{r}{6a_0}\right)\left(\frac{r}{a_0}\right) e^{(-r/3a_0)}$